

Program 1: Mutex using POSIX Threads

```
GNU nano 7.2
#include <stdio.h>
#include <pthread.h>

int count = 0;
pthread_mutex_t lock;

void* increment(void* arg)
{
    for(int i = 0; i < 100000; i++)
    {
        pthread_mutex_lock(&lock);
        count++;
        pthread_mutex_unlock(&lock);
    }
    return NULL;
}

int main()
{
    pthread_t t1, t2;

    pthread_mutex_init(&lock, NULL);

    pthread_create(&t1, NULL, increment, NULL);
    pthread_create(&t2, NULL, increment, NULL);

    pthread_join(t1, NULL);
    pthread_join(t2, NULL);

    printf("Final count = %d\n", count);

    pthread_mutex_destroy(&lock);

    return 0;
}
```

```
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical/Mmap_Practical/Ext_Practical/Mutex_Practical$ nano mutex.c
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical/Mmap_Practical/Ext_Practical/Mutex_Practical$ gcc mutex.c -o mutex -pthread
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical/Mmap_Practical/Ext_Practical/Mutex_Practical$ ./mutex
Final count = 200000
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical/Mmap_Practical/Ext_Practical/Mutex_Practical$ |
```

Program 2: Condition Variables (Producer-Consumer)

```
GNU nano 7.2
#include <stdio.h>
#include <pthread.h>

pthread_mutex_t mutex = PTHREAD_MUTEX_INITIALIZER;
pthread_cond_t cond = PTHREAD_COND_INITIALIZER;

int data = 0;
int ready = 0;

void* producer(void* arg)
{
    pthread_mutex_lock(&mutex);

    data = 100;
    ready = 1;

    printf("Producer produced: %d\n", data);

    pthread_cond_signal(&cond);
    pthread_mutex_unlock(&mutex);

    return NULL;
}

void* consumer(void* arg)
{
    pthread_mutex_lock(&mutex);

    while(ready == 0)
        pthread_cond_wait(&cond, &mutex);

    printf("Consumer consumed: %d\n", data);

    pthread_mutex_unlock(&mutex);

    return NULL;
}

int main()
{
    pthread_t p, c;

    pthread_create(&c, NULL, consumer, NULL);
    pthread_create(&p, NULL, producer, NULL);

    pthread_join(p, NULL);
    pthread_join(c, NULL);

    pthread_mutex_destroy(&mutex);
    pthread_cond_destroy(&cond);

    return 0;
}
```

```
Final Count = 200000
sairedy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical/Mmap_Practical/Ext_Practical/Mutex_Practical$ nano producer.c
sairedy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical/Mmap_Practical/Ext_Practical/Mutex_Practical$ gcc producer.c -o producer -pthread
sairedy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical/Mmap_Practical/Ext_Practical/Mutex_Practical$ ./producer
Producer produced: 100
Consumer consumed: 100
sairedy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical/Mmap_Practical/Ext_Practical/Mutex_Practical$ |
```